

A COVINDEX based on a GAM beta regression model with an application to the COVID-19 pandemic in Italy

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Abstract Detecting changes in COVID-19 disease transmission over time is a key indicator of epidemic growth. Near real-time monitoring of the pandemic growth is crucial for policy makers and public health officials who need to make informed decisions about whether to enforce lockdowns or allow certain activities. The effective reproduction number R_t is the standard index used in many countries for this goal. However, it is known that due to the delays between infection and case registration, its use for decision making is somewhat limited. In this paper a near real-time COVINDEX is proposed for monitoring the evolution of the pandemic. The index is computed from predictions obtained from a GAM beta regression for modelling the test positive rate as a function of time. The proposal is illustrated using data on COVID-19 pandemic in Italy and compared with R_t . A simple chart is also proposed for monitoring local and national outbreaks by policy makers and public health officials.

Keywords pandemic surveillance; GAM beta regression; COVINDEX; public-health decision-making

1. Introduction

The World Health Organization (WHO) declared coronavirus disease (COVID-19) a pandemic on 11 March 2020. Since then, most countries around the world have addressed this threat by implementing various strategies to fight the pandemic. From simple preventive measures, such as case identification and contact tracing, quarantine and isolation, to more severe strategies based on general lockdowns of all non-essential economical and social activities. Since public health decision-making requires the balancing of numerous, and often conflicting, factors, a timely and data-informed decision making process appears crucial.

The basic reproduction number, R_0 , is an indicator of the epidemic's virulence. It is defined as the average number of infections caused by an infected person when the whole population is susceptible, and for SARS-CoV-2 is between 2 and 3 (Li et al., 2020; Hilton and Keeling, 2020). As the pandemic evolves, the effective reproduction number R_t is a more useful measure. This is the average number of infections that an infected person will cause. An R_t above 1.0 indicates that the outbreak is growing, and below 1.0 means that it is shrinking. As a simple understood measure, R_t is publicly reported every week, and it has been used in many countries, including Italy, to decide whether to tighten or loosen control measures. However, R_t suffers from several drawbacks when used to monitor the transmission of the disease over time, the main one being the delay with which it signals the evolution of the pandemic (Gostic et al., 2020; Adam, 2020). Therefore, with a delay on the estimate of R_t between ten days to two weeks, the use of R_t as a near real-time decision-making tool appears rather pointless.

This paper introduces a COVID-19 index, called COVINDEX, which tries to assess whether the epidemic is growing, shrinking, or holding steady. The proposed index is estimated by modelling the percent positive rate (PPR), also called test positive rate (TPR), with a GAM beta regression model. TPR is an easily computed statistic, defined as the percentage of all COVID-19 tests performed on a given day that are actually positive. This metric can be used to understand the spread of the virus, but it also offers a measure of how adequately a country is testing. TPR can be high if the number of positive tests is too high, but also if the number of total tests is too low. Most developed countries faced limited testing capacity during the initial phase of the pandemic, which resulted in high TPR values due to testing conducted primarily on symptomatic individuals. In the following months the ability to administer tests using PCR (polymerase chain reaction) or molecular swabs largely

increased, leading to a situation that allows both symptomatic and asymptomatic individuals to be tested. Although TPR can't be used for estimating incidence of the virus in the general population, a fundamental epidemic parameter that would require a carefully designed sampling plan, it can be used for monitoring the evolution of infection and transmission in the community. Higher positive rates suggest the need for further restrictions, such as wearing masks and physical distancing, to slow down the spread of the disease. As a rule of thumb, World Health Organization recommended 5% as the threshold for the percent positive rate to declare the COVID-19 transmission under control.

The main advantages of the proposed COVINDEX is the use of data routinely collected and its timely estimation which provides a near real-time tool to assess the effectiveness of interventions and to inform policy. Furthermore, since it is based on a statistical model, the associated uncertainty can be estimated.

The paper is organized as follows. Section 2 introduces the GAM beta regression model, its estimation and uncertainty assessment. Section 3 describes the proposed COVINDEX and its usage for monitoring the pandemic evolution. Section 4 includes a detailed analysis of COVID-19 pandemic in Italy, including the estimation of COVINDEX, from early March 2020 to the end of March 2021. Section 5 contains a comparison between the proposed COVINDEX and the effective reproduction number, showing the advantages of COVINDEX as near real-time monitoring tool. The final section provides some concluding remarks.

2. Statistical Model for the Test Positive Rate

2.1. GAM beta regression

Let y_t be the test positive rate (TPR), defined as the ratio of the number of new positive cases P_t to the number of tests T_t at time t . As a proportion TPR is naturally limited in the range $[0, 1]$. Several approaches and models can be used for response variables that are expressed as proportions (Douma and Weedon, 2019), and perhaps the most popular statistical model is the beta regression model (Ferrari and Cribari-Neto, 2004; Zeileis and Cribari-Neto, 2010).

Assume that TPR can be modelled by a beta distribution written as

$$y_t \sim \text{Beta}(\mu_t, \phi),$$

with mean and variance of the beta distribution given, respectively, by

$$\mathbb{E}[y_t] = \mu_t,$$

and

$$\mathbb{V}[y_t] = \frac{\mu_t(1-\mu_t)}{1+\phi}.$$

The mean μ_t can be expressed as a function of the linear predictor $\eta_t = \beta^\top \mathbf{x}_t$, where β is a vector of unknown regression coefficients, and $\mathbf{x}_t$ is the vector of observed values on, say, p predictors. Usually, the logistic function is used in beta regression, so we can write

$$\mu_t = \text{logistic}(\eta_t) = \frac{\exp(\eta_t)}{1 + \exp(\eta_t)} = \frac{1}{1 + \exp(-\eta_t)}.$$

The inverse of the logistic function is the logit function, the so-called *link* function in GLM terminology (McCullagh and Nelder, 1989), given by:

$$\text{logit}(\mu_t) = \log\left(\frac{\mu_t}{1-\mu_t}\right) = \eta_t.$$

Generalized Additive Models (GAMs; Hastie and Tibshirani, 1990) allows to model the dependence of the response variable in a flexible way using smooth functions of the predictors by defining the linear predictor as

$$\eta_t = \beta_0 + \sum_{j=1}^p f_j(x_{tj}),$$

where $f_j(x_{tj}) = \sum_{k=1}^K \beta_{jk} B_{jk}(x_{tj})$ is the smoothing term for the j th predictor with $\{B_{jk}()\}_{k=1}^K$ a set of known basis functions associated to unknown parameters β_{jk} . Several smoothers can be defined by adopting different basis functions, such as penalized regression splines, cubic regression splines, and thin plate regression spline. For an overview of the several smoothing functions available using spline bases see Wood (2017, Chapter 5).

In our application the only feature included in the linear predictor is time, so x_t is an integer counting the days since January 1st, 2020. To some extent, the coding of such feature has no practical consequence, and other equivalent forms could have been used as well. Thus, in our case the linear predictor of GAM simplifies to

$$\eta_t = \beta_0 + \sum_{k=1}^K \beta_k B_k(x_t).$$

Thin plate regression splines are used for the basis $\{B_k\}_{k=1}^K$. These produce penalized reduced rank smooth terms without the need to specify the knots. Note, however, that other smooth functions would have given nearly equivalent results.

2.2. Estimation

Estimation of the GAM model introduced in previous section can be pursued by REstricted Maximum Likelihood (REML), which amounts to maximize the penalized log-likelihood

$$\ell_p(\beta) = \ell(\beta) - \frac{1}{2} \lambda \beta^T S \beta. \quad (1)$$

The last term in the right-hand side represents the smoothing penalty, with λ a smoothing parameter, and S a known penalty matrix. The selection of the smoothing parameter can be obtained, among many other proposals, by minimizing the Akaike's information criterion (AIC).

However, because the number of administered swabs is not constant over time, we must take into account this fact when modelling the test positive rate. There are several reasons for this empirical evidence. First of all, during the weekends (especially on Sundays) the number of swabs drops drastically. Furthermore, during periods of strong expansion of the pandemic, the monitoring system is unable to carry out effective surveillance and only symptomatic patients are likely to be tested. Accounting for the different number of swabs in the model for the positive rate can be achieved by adopting a weighted penalized log-likelihood criterion. This amounts to replace the log-likelihood $\ell(\beta)$ in (1) with the weighted version

$$\ell_w(\beta) = \sum_{t=1}^n w_t \ell(y_t | \beta),$$

where w_t are prior weights specifying the contribution of each data point to the log-likelihood. In particular, indicating with $\bar{T}$ the average number of administered swabs over the period, weights can be defined as $w_t = T_t / \bar{T}$ so that positive rates y_t computed from number of swabs larger than the average have proportionally larger weights, and vice versa for those rates based on number of swabs smaller than the average. Furthermore, with the adopted definition for the weights the contribution of each datum is specified without changing the overall magnitude of the log-likelihood.

Once the model is fitted, the predicted TPR can be computed as

$$\hat{\mu}_t = \text{logistic} \left(\hat{\beta}_0 + \sum_{k=1}^K \hat{\beta}_k B_k(x_t) \right). \quad (2)$$

2.3. Uncertainty and inference

The penalized likelihood approach described above has also a Bayesian interpretation by assuming an improper multivariate normal prior on β . In this case, the REML estimates of β coefficients are also the MAP of the Bayesian posterior distribution, with the latter given by

$$\beta|(y, \lambda) \sim N(\hat{\beta}, (\hat{I} + \lambda S)^{-1}), \quad (3)$$

where $\hat{I}$ is the observed information matrix (Hessian of the negative log-likelihood) at $\hat{\beta}$ (Wood, 2017, Section 6.10). This result is useful for computing approximate credible intervals for any function of β by simulating from the posterior (Gelman and Hill, 2006, Section 7.2). Wood (2017, p. 294) reported good frequentist coverage properties for such Bayesian credible intervals, with empirical coverage close to the nominal level when averaged across the domain of the function.

In practice, coefficients β^* are simulated from (3), and then plugged in equation (2) to get the simulated means μ_t^* . The process is replicated a large number of times, say 10 000 or more, and the percentiles of the simulated distributions at different values of x_t can be used to compute the limits of approximate credible intervals.

3. COVINDEX as a Monitoring and Decision-Making Tool

The COVINDEX proposed in this paper is an attempt to compute a synthetic index summarizing the evolution of the COVID-19 pandemic, which can be useful to policy makers and public health officials for monitoring local and national outbreaks. In our proposal this is simply computed as

$$\text{COVINDEX}_t = \frac{\hat{\mu}_t}{\hat{\mu}_{t-7}}, \quad (4)$$

the ratio of the predicted positive rate at time t to the prediction 7 days earlier. The value of 7 is chosen because it is approximately the expected incubation time for COVID-19 (Nazar and Elfadil, 2021), and because it corresponds to the observed weekly fluctuation in testing. A COVINDEX value larger than 1.0 means that the pandemic is growing, while a value smaller than 1.0 indicates that new infections are slowing down. Of course, uncertainty also affects the COVINDEX and the approach outlined in Section 2.3 for the test positive rate can be used here as well. In particular, from each simulated series of values μ_t^* , the simulated COVINDEX series can be obtained by applying equation (4), and approximate credible intervals can be computed from the percentiles of the simulated distribution.

We argue that decisions made by policy makers should be based both on the COVINDEX, which provides an outlook on the likely behaviour of the pandemic in the near future, and on the level of the estimated TPR, which represents its current status. Following this idea, a TPR-COVINDEX risk quadrant chart can be drawn (see Figure 1). This chart illustrates four potential scenarios which represent a useful tool for a decision maker. The quadrants are defined by the dashed lines drawn at selected threshold values. For COVINDEX the natural reference value is 1.0, with values below it indicating a shrinking outbreak, and values higher than 1.0 indicating epidemic situations that are increasingly worrying and out of control. Note that, since the index is a ratio, the y-axis is expressed in logarithmic scale. For the positive rate, the threshold value can be set according to the World Health Organization, which published a set of criteria to inform whether the epidemic is under control. In particular, one criterion states that “[...] less than 5% of samples positive for COVID-19, at least for the last 2 weeks, assuming that surveillance for suspected cases is comprehensive” (World Health Organization, 2019).

According to the above mentioned threshold values, the upper-right quadrant represents the worst-case scenario, with high values of both TPR and COVINDEX. On the contrary, the best-case scenario is the lower-left quadrant which has both low TPR and COVINDEX less than 1.0 indicating a decreasing circulation of the virus. The remaining quadrants are intermediate cases. Typical situations will move in a clockwise direction, moving from the worst-case, represented by the red

quadrant on top-right, to the orange quadrant at bottom-right, and eventually reaching the yellow quadrant indicating an outbreak under control. However, in some cases the pandemic could regain strength by getting COVINDEX values greater than 1.0, thus moving towards the top-left orange quadrant or directly towards the worst-case situation described by the red quadrant. A description of the Italian situation since March 2020 is discussed in Section 4.

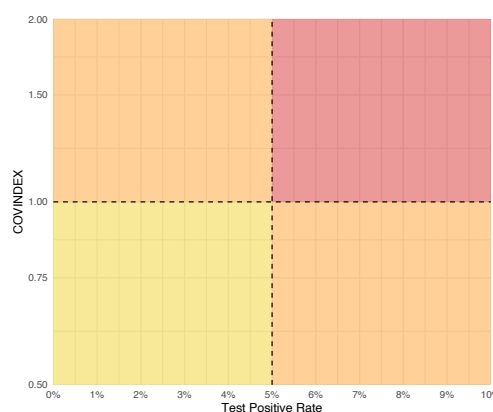


Figure 1. TPR-COVINDEX risk quadrant chart.

4. Application to Italian COVID-19 Pandemic

4.1. Data

The Italian Department of Protezione Civile provides daily information on the COVID-19 pandemic, both at the national and the regional level, in a public GitHub repository (Presidenza del Consiglio dei Ministri – Dipartimento della Protezione Civile, 2020). Among the data contained in this repository, the cumulative number of naso-pharyngeal or molecular swabs and the corresponding positive tests are provided. Starting with January 15th, 2021, antigen tests are also officially recorded, while previously only some regions included them in the recorded statistics since autumn 2020. The reliability of such information is at best questionable and not available uniformly for the year 2020. For these reasons, **in our analyses we only considered the information from molecular swabs to compute the test positive rate (TPR)**, a commonly used screening and diagnostic tool for COVID-19 (World Health Organization, 2020). The plot on **Figure 2 shows the observed TPR over time** with points proportional to the administered swabs.

4.2. COVINDEX estimate

Table 1 reports the summary output of the estimated beta GAM regression model for the test positive rate in Italy from early March 2020 to end of March 2021. The amount of smoothing applied to the time predictor is selected by AIC.

Table 1. GAM beta regression model summary

Num. of obs. = 392		Dispersion par. = 871.53		
Log-likelihood = 1360.2		Deviance expl. = 0.9839		
REML = -1285.7		AIC = -2662.4		
Parametric coefficients:				
	Estimate	Std. error	z-value	p-value
(Intercept)	-3.119	0.01672	-186.6	< 0.001
Smooth terms:				
	edf	Ref. df	chi.sq-value	p-value
s(t)	26.7	30.32	8034	< 0.001

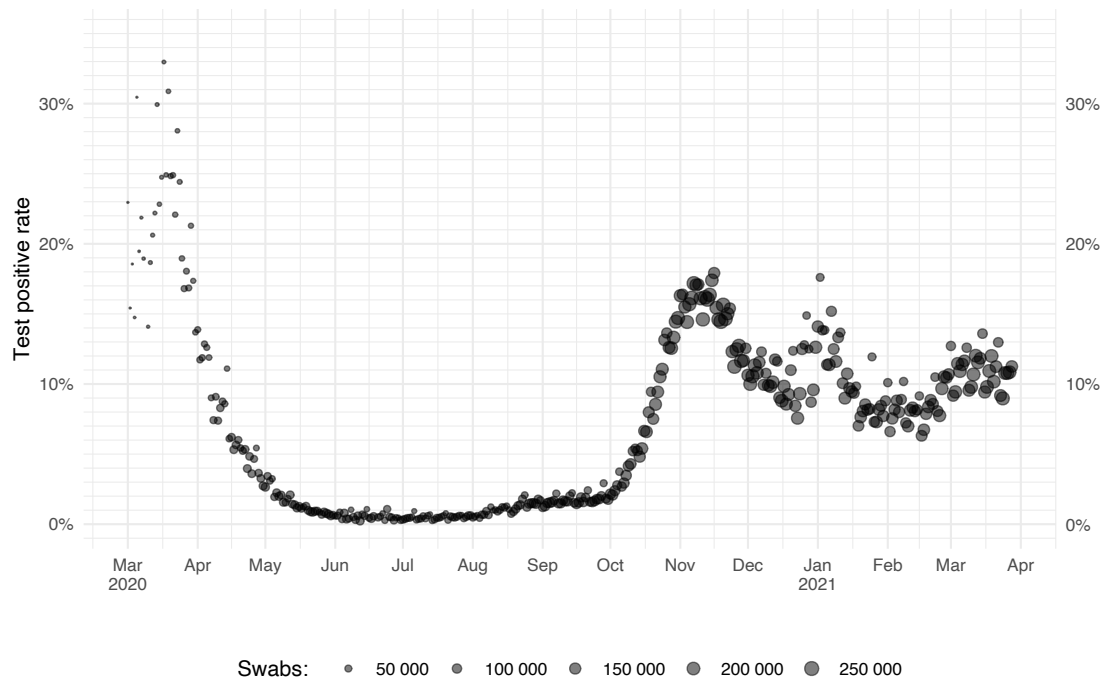


Figure 2. Plot of test positive rate from beginning of COVID-19 pandemic in Italy to the end of observational period with size of points proportional to the number of molecular swabs administered.

Figure 3 reports the estimated curve for the test positive rate with 95% credible intervals for the mean obtained by simulating from the posterior distribution as described in Section 2.3. The highest positive rates are achieved in March 2020 during the first wave of pandemic, and on November 2020, corresponding to the second wave. A resurgence of spread during the end of 2020 is followed by a quick decrease in earlier 2021. Subsequently, the situation remained stable for about a month, but in the second part of February another sharpe increase occurred due to the appearance of COVID-19 variants in the Italian territory.

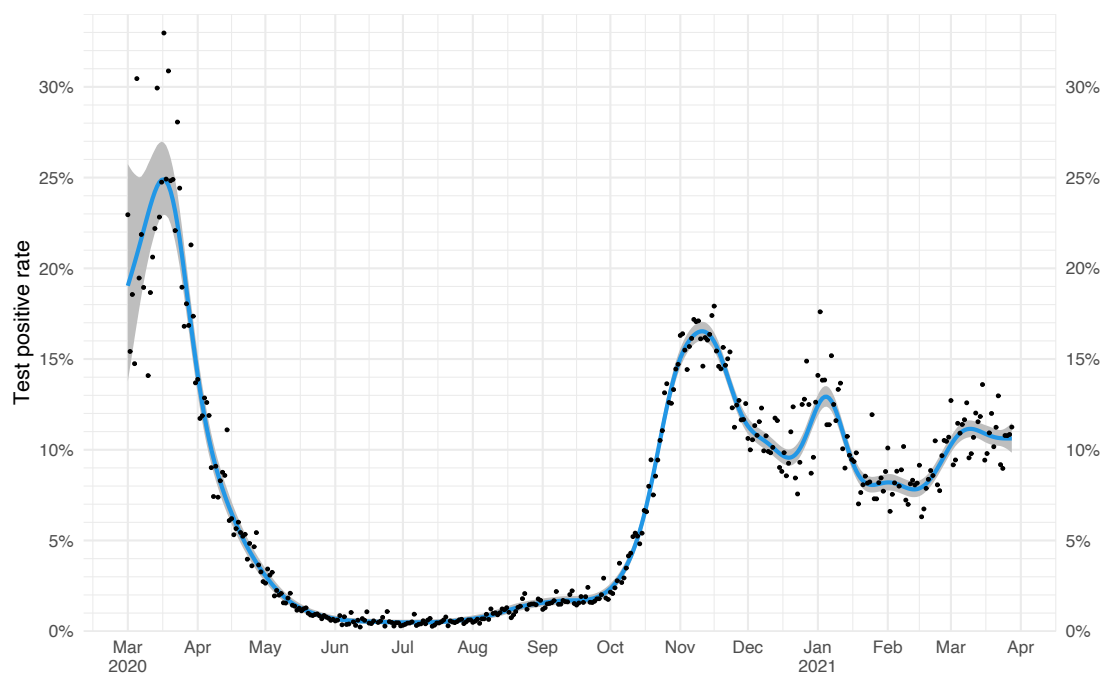


Figure 3. GAM estimate of test positive rate in Italy during 2020 and earlier months of 2021, with 95% simulated credible intervals for the mean.

Based on the estimated model and uncertainty for the test positive rate, the COVINDEX is computed following equation (4). Figure 4 shows the estimated COVINDEX with 95% credible intervals. Notice that the y-axis is expressed on logarithmic scale, the natural scale to visualize ratios (Wilke, 2019, , Sec. 3.2). The index fluctuates widely throughout the year 2020, following the periods of expansion and contraction of the spread of the pandemic. After the first wave in spring 2020 we observe a quick decreasing trend, followed by a slowly increase during the summer, corresponding to a relaxation of the containment measures, with values significantly larger than 1.0 during August. This represents the first signal of a resurgence of the pandemic. Sharpe and large increases are also observed during October in conjunction with the second wave that strongly affected Italy. Two additional peaks are detected at the end of 2020 and on February 2021, corresponding to gradual relaxation of containment measures before Christmas and mid January, with the latter that occurred during the period of political instability associated with the change of government.

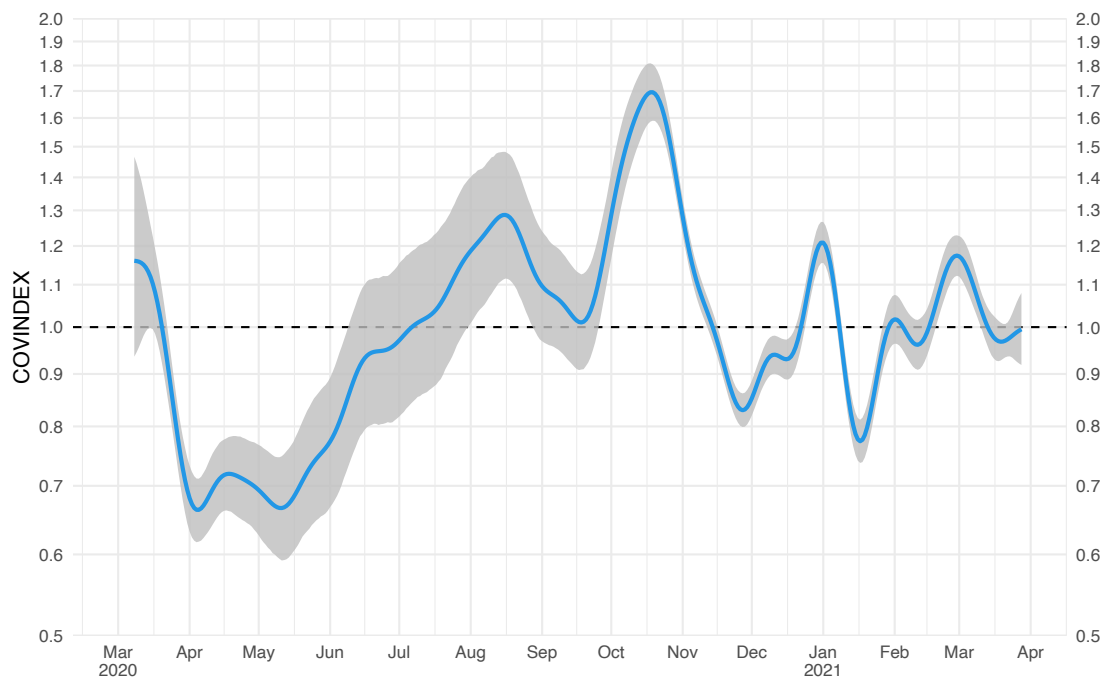


Figure 4. Evolution of COVINDEX (on logarithmic scale) for Italy during 2020 and earlier months of 2021, with approximate 95% credible intervals.

4.3. TPR-COVINDEX risk quadrant chart analysis

Figure 5 shows the TPR-COVINDEX risk quadrant chart for Italy, with points connected following the temporal path, a graph also known as connected scatterplot (Haroz et al., 2015). The curve concisely represents the evolution of both indices during the pandemic. Starting with the critical situation in March 2020, the situation improved in the following months, moving from the red quadrant to the orange bottom-right quadrant and then the yellow quadrant during summer 2020. By the end of summer 2020 we observe a worsening of the situation that lead to the red quadrant in November. In the following months there has been a constant oscillation between the red and right-orange quadrants, indicating a serious pandemic situation.

A scatterplot of TPR vs COVINDEX is also useful for surveillance of the pandemic in different Italian regions. Figure 6 summarizes the status of the pandemic for the Italian regions at selected time points. A high-risk situation is observed at the beginning of November 2020, where all regions belong to the red quadrant. The following month saw an improvement with most regions moving towards the bottom-right orange quadrant. A more complex and varied situation is observed between February and March 2021, with some regions moving from the red to the orange quadrant, and vice versa for other regions.

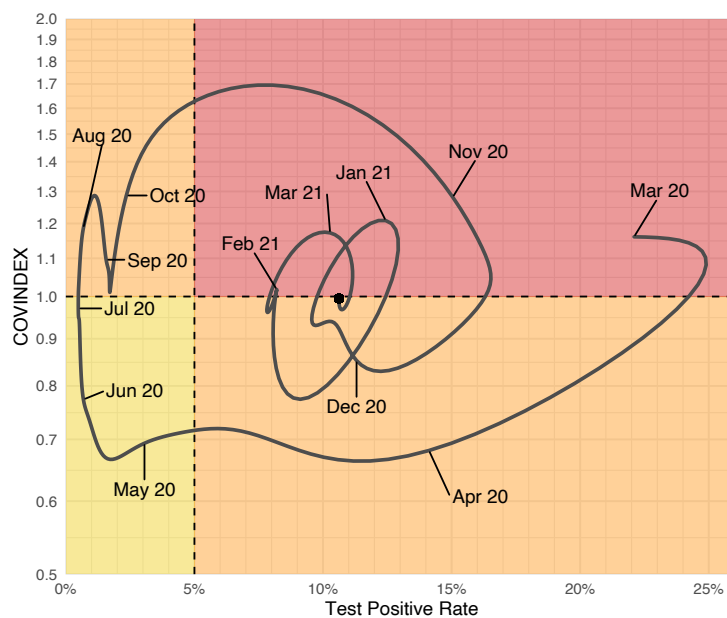


Figure 5. TPR-COVINDEX risk chart as connected scatterplot for Italy. The first day of each month is highlighted to provide a temporal reference.

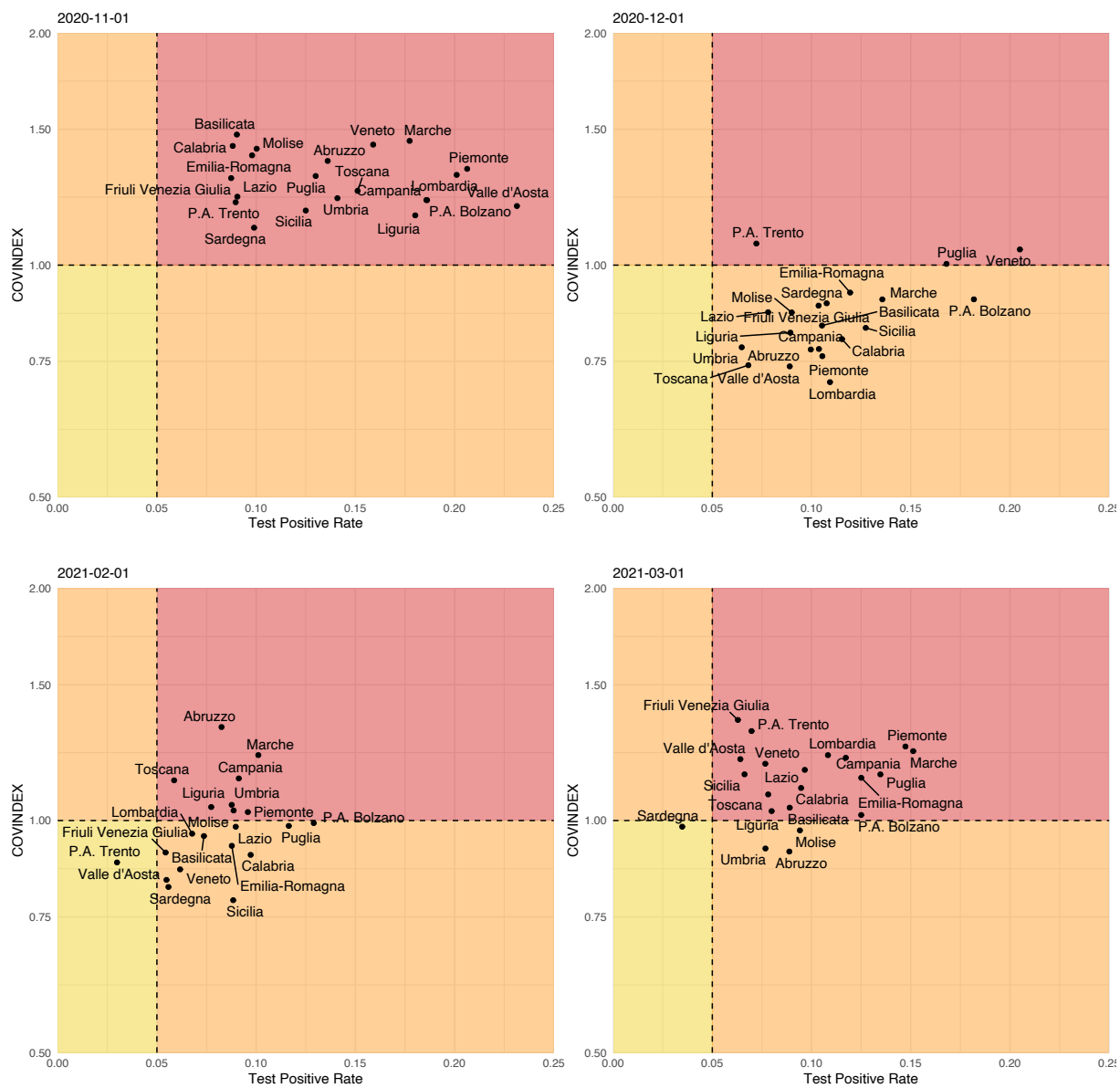


Figure 6. TPR-COVINDEX plot for Italian regions at different time points.

5. A comparison of COVINDEX with the effective reproduction number

The main index used in Italy for pandemic surveillance is R_t , the effective reproduction number. The procedure employed for estimating R_t is described by Guzzetta and Merler (2020) and it is based on the Bayesian methodology of Cori et al. (2013). Details can be found at <https://www.epicentro.iss.it/coronavirus/sars-cov-2-sorveglianza-dati>. An archive containing both the data and the R script used by the Italian National Institute of Health (ISS) for computing R_t is available at https://www.epicentro.iss.it/coronavirus/open-data/calcolo_rt_italia.zip.

In this Section we provide a comparison of the proposed COVINDEX with the values of R_t estimated following the procedure outlined above for Italy from March 2020 to March 2021. Furthermore, since the effective reproduction number does not provide a timely snapshot of the evolution of the pandemic, we also provide here two examples showing the failure of R_t to highlight the likely evolution of the pandemic and we compare its behaviour with the proposed COVINDEX.

The top graph reported in Figure 7 shows the estimated curves for COVINDEX and R_t . Overall, a similar trend can be observed for the two curves, particularly since October 2020. R_t appears to be more wiggly than COVINDEX, especially during the summer 2020. Likely, this is related to the large uncertainty in that period due to the relative small number of positive cases (around few hundreds) observed in that period.

As mentioned in Section 4.2, the second wave of COVID-19 epidemic hit Italy between the second half of October and the beginning of November 2020, followed by a rapid decrease during the remainder of the month. However, from the beginning of December 2020 it was evident that this decline had stopped and that the situation was starting to get worse. This is clearly indicated by the upward slope of the COVINDEX computed on December 5th and shown in the bottom-left graph of Figure 7. On the contrary, the R_t index calculated on the same day, with estimates considered valid up to 14 days before, produces a curve which erroneously suggests a decline in the spread of the pandemic. However, if the R_t curve is estimated a week later, we begin to see an increase in the spread of the pandemic (see the dotted red curve in bottom-left graph of Figure 7). The main problem is that such alert is reported too late.

A similar situation is also faced at the beginning of March 2021. After a period of almost constant positive rate during February 2021, with both COVINDEX and R_t oscillating around 1.0, by the end of the month there was a clear increase of the test positive rate. This was immediately signalled by COVINDEX computed on February 28th 2021 (see bottom-right graph in Figure 7), but R_t computed on the same day was still signalling a steady state and only after a week an increasing value of R_t would have signalled the resurgence of the pandemic.

6. Final comments

In this paper we have proposed an index, named COVINDEX, that can be used for near real-time monitoring of COVID-19 pandemic. The index is computed as the ratio of the estimated test positive rate on a given day with respect to the value estimated for a week before. Estimation of test positive rates is obtained by statistical modelling the daily empirical positive rates calculated from the observed data. To this end, a GAM beta regression model with weights proportional to the administered tests is fitted. By exploiting the relationship of penalized likelihood for GAMs with MAP Bayesian estimation, credible intervals for COVINDEX can be obtained via simulation to express the associated uncertainty.

We applied the proposed methodology to the Italian COVID-19 outbreak and we compared the trend of COVINDEX to the effective reproduction number R_t . The analyses carried out confirm that R_t is a delayed index of epidemic trend, and for this reason may provide a biased picture of the current pandemic status. On the contrary, COVINDEX seems to provide a more up-to-date information which can be used as a decision-making tool. This aspect is of crucial importance for all policy makers and public health officials.

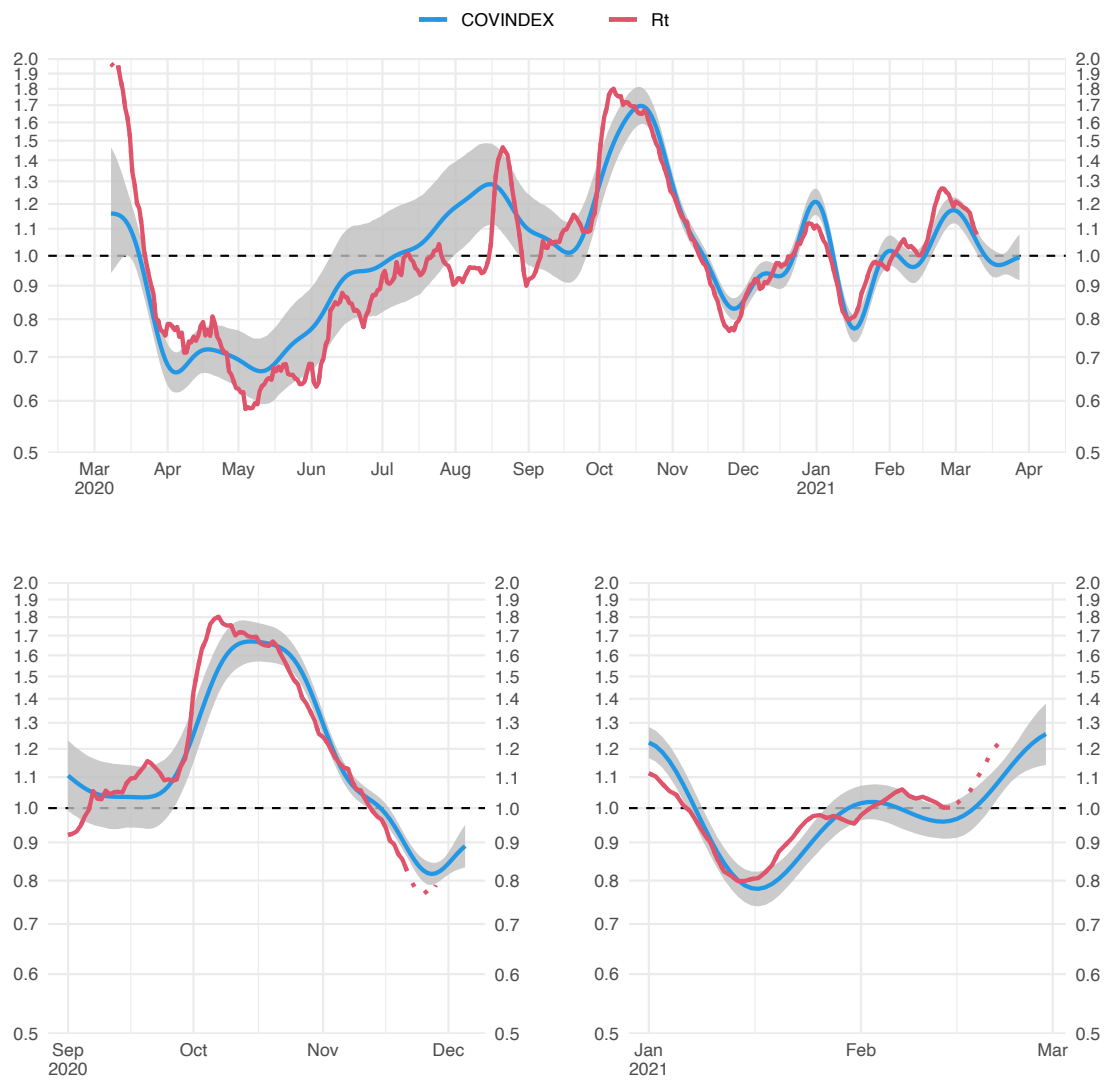


Figure 7. Comparison of COVINDEX and R_t for Italy. Bottom panels show the comparison at early December 2020 (left) and at the end of February 2021 (right).

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